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Dai et al.

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(54) **STRAWBERRY PLANT NAMED ‘TAMIR’**

(50) Latin Name: *Fragaria ananassa*
Varietal Denomination: **TAMIR**

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(52) **U.S. Cl.** **Plt./209**

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Plt./209
See application file for complete search history.

(56) **References Cited**

OTHER PUBLICATIONS

Kunihisa, et al., “Development of Cleavage Amplified Polymorphic Sequence (CAPS) Markers for Identification of Strawberry Cultivars,” *Euphytica*, 2003, vol. 134, No. 2, pp. 209-215.

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of strawberry plant named ‘TAMIR’. This new strawberry plant named ‘TAMIR’ is primarily adapted to the climate and growing conditions of the growing conditions of the Sharon coastal plain, at 32° latitude, located in Israel, and is primarily characterized by low amounts of runners (stolons) production in the propagating nursery; no chilling requirement for flower induction; very early fruit production from the first week of November in Israel; moderate plant size with open, flattened habit; fruiting trusses position level with foliage; large fruit size, orange-red in color; high fruit firmness even in high temperatures; juicy fruit texture with strong flavor; very nice appearance with yellowish achenes over bright red fruit skin; and high field tolerance to powdery mildew.

7 Drawing Sheets

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Latin name of the genus and species of the plant claimed:
Fragaria ananassa.

Variety denomination: ‘TAMIR’.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct strawberry variety named ‘TAMIR’. The variety is botanically known as *Fragaria ananassa* Duch.

The new strawberry ‘TAMIR’ is a product of a planned breeding program conducted by the inventors, Nir Dai, Zecharia Tanami and Sara Slotzky in Bet-Dagan, Israel. The objective of the breeding program is to develop a new infra short-day strawberry variety having premium fruit quality (appearance, firmness and flavor), which produces high marketable yields from the beginning of November.

This new strawberry ‘TAMIR’ is a result of a controlled cross made by the inventors in 2000, in Bet-Dagan, Israel. The female or seed parent is strawberry variety designated ‘HADAS’, registered for PBR in Israel (No. 1860, Date: Jan. 20, 1999) and in Europe (No. EU 6445, Date: Jul. 3, 2000). The male or pollen parent is strawberry variety designated ‘HERUT’, registered for PBR in Israel (No. 2243, Date: May 2, 2002) and in Europe (No. EU 21583, Date: Dec. 17, 2007). The new strawberry ‘TAMIR’ was discovered and selected by the inventors as a single flowering plant within the progeny of the stated cross in March of 2001, in controlled environment of an experimental greenhouse in Bet-Dagan, Israel. After its selection, the new variety was asexually propagated by sto-

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lons in a nursery located in Bet-Dagan, Israel. The new variety was extensively tested over the following years in few small commercial fields in Moshav Tsofit, Kadima and Qalansawa, Israel. This propagation has demonstrated that the combination of characteristics as herein disclosed for the new cultivar are firmly fixed and retained through successive generations of asexual reproduction. The new cultivar reproduces true to type.

BRIEF SUMMARY OF THE INVENTION

‘TAMIR’ is primarily adapted to the climate and growing conditions of the Sharon coastal plain, at 32° latitude, located in Israel and to similar climatic regions. This region provides the needed growing conditions of autumn with high temperatures that promote primary plant growth and earliness of fruit production; and mild winter with moderate temperatures and many sunny days with low humidity, maintaining plant vigorous and fruit quality during the production months.

The following traits have been repeatedly observed and are determined to be unique characteristics of ‘TAMIR’, which in combination distinguish this strawberry plant as a new and distinct variety:

1. Low amounts of runners (stolons) production in the propagating nursery;
2. No chilling requirement for flower induction;
3. Very early fruit production from the first week of November in Israel;
4. Moderate plant size with open, flattened habit;

- 5. Fruiting trusses position level with foliage;
- 6. Large fruit size, orange-red in color;
- 7. High fruit firmness even in high temperatures;
- 8. Juicy fruit texture with strong flavor;
- 9. Very nice appearance with yellowish achenes over bright red fruit skin; and
- 10. High field tolerance to powdery mildew.

Plants of the new strawberry variety ‘TAMIR’ differ from plants of the parents, ‘HADAS’, registered for PBR in Israel (No. 1860, Date: Jan. 20, 1999) and in Europe (No. EU 6445, Date: Jul. 3, 2000), and ‘HERUT’, registered for PBR in Israel (No. 2243, Date: May 2, 2002) and in Europe (No. EU 21583, Date: Dec. 17, 2007) in the characteristics described in Table 1.

TABLE 1

Characteristic	‘TAMIR’	‘HADAS’ Registered	‘HERUT’ Registered
Average fruit weight (gm)	22 gram	25 gram	20 gram
Unevenness of fruit surface	Absent or very weak	Weak	Weak
Insertion of achenes	Level with surface	Level with surface	Slightly below surface
Attitude of calyx	Clasping or detached	Clasping or detached	Clasping or detached
Firmness of the fruit	Firm	Medium	Medium
Plant size	Medium	Medium	Large
Average number of stolons	Few	Many	Very many
Foliage color (upper surface)	Medium to dark	Dark	Medium to dark
Interveinal leaf blistering	Weak	Weak	Weak
Vigor	Medium	Medium	Strong
Position relative to foliage	Level with	Above	Beneath

Of the many commercial cultivars known to the present inventor, the most similar in comparison to the new strawberry variety ‘TAMIR’ is the male or pollen parental cultivar, strawberry variety ‘HERUT’ (registered for PBR in Israel and Europe). Plants of the new strawberry variety ‘TAMIR’ differ from plants of strawberry variety ‘HERUT’ in the characteristics described in Table 2.

TABLE 2

Characteristic	‘TAMIR’	‘HERUT’ Registered
Average fruit weight (gm)	22 gram	20 gram
Insertion of achenes	Level with surface	Slightly below surface
Attitude of calyx	Clasping or detached	Clasping or detached
Firmness of the fruit	Firm	Medium
Plant size	Medium	Large
Average number of stolons	Few	Very many
Fruit color	Orange-red	Red to dark red
Fruit tip color	Without green tip	Dominant green tip
Foliage size	Medium	Large
Inflorescence position relative to foliage	Level with	Beneath

For its identification, a series of molecular markers have been developed for this new variety. The genetic polymorphic pattern of two Cleavage Amplified Polymorphic Sequence (CAPS) markers (A. APX- MluI and B. CTI-HinI (Kunihisa et al., 2003)) that differ between the new strawberry cultivar

‘TAMIR’ and the main commercial Israeli strawberry cultivars, as well as, other widely known strawberry cultivars is shown in FIG. 7.

BRIEF DESCRIPTIONS OF THE PHOTOGRAPHS

The accompanying color photographs illustrate the overall appearance of typical specimens of the new strawberry variety ‘TAMIR’, at various stages of development as true as it is reasonably possible with color reproductions of this type. Color in the photographs may differ slightly from the color value cited in the botanical description which accurately describes the color of ‘TAMIR’. The depicted plant and plant parts of the new strawberry variety ‘TAMIR’ were taken in Moshav Tsofit, Israel, and are approximately 5 month old.

FIG. 1 shows typical fruiting field characteristics of ‘TAMIR’ taken in February of 2007.

FIG. 2 shows a close-up view of the typical leaf structure of ‘TAMIR’ taken in February of 2008.

FIG. 3A shows typical mature and immature field fruit of ‘TAMIR’ taken in February of 2007 and

FIG. 3B shows typical mature and immature field fruit of ‘TAMIR’ taken in February of 2008.

FIG. 4 shows a close-up external view of mature fruit of ‘TAMIR’ taken in February of 2007.

FIG. 5 shows a close-up internal view of mature fruit characteristics of ‘TAMIR’ taken in February of 2008.

FIG. 6. Average August to March maximum and minimum temperatures, during the main strawberry growing season at the fruit production field.

FIG. 7. The genetic polymorphic pattern of 12 strawberry accessions at two CAPS markers: A. APX- MluI and B. CTI-HinI (following Kunisha, et al. (2003)).

DETAILED BOTANICAL DESCRIPTION

‘TAMIR’ has not been observed under all possible environmental conditions. The characteristics of the new variety may vary in detail, depending upon variations in environmental factors, including weather (temperature, humidity and light intensity), day length, soil type and location.

The aforementioned photographs, together with the following observations, measurements and values describe the new strawberry variety ‘TAMIR’, unless otherwise noted, taken during the winter of the growing season 2007–2008 in Moshav Geolim and Bet-Dagan, Israel. The observations, measurements and values were taken from plants of ‘TAMIR’ dug from a low-elevation nursery located in Moshav Geolim and Bet-Dagan, Israel, during September, 2007 and planted directly, without any chilling, in the same day in the commercial field in Moshav Geolim and experimental greenhouse Bet-Dagan, Israel. Plants of the new strawberry variety ‘TAMIR’ were grown under conditions which closely approximate those generally used in commercial practice.

Growing conditions at the nursery:

Mother plants provided from an authorized nursery are planted in the beginning of May, in the field nursery at 2.5 m×2.5 m distance. Combination of overhead and drip irrigation with addition of fertilizers is used. The average day/night temperatures during the establishment of the daughter plants, between June to August, are 32° C./18° C. respectively. Runners appear from June, which produce young daughter plants up till mid September, these are then collected and planted directly in the commercial fields.

Growing conditions at the fruit production field:

The bare rooted plants are washed from soil and dipped in Octav (Prochloraz manganese) 2 g/Liter solution to avoid anthracnose. These bare rooted plants are then planted in raised beds with overhead irrigation in 4 rows at 30 cm apart. Distance between plants within rows is 32 cm. During the first month, no fertilizers are added, due to the addition of 60–100 m³/hectar compost during field preparation. About 4 weeks after planting the beds are covered with silvery-black, 30 micron polyethylene. The young plants are pulled out through the pre-prepared holes. From this point on, the plants are irrigated with a drip system with a supply of additional fertilizers, as is typically used in agricultural practice. By the end of October the beds are covered with transparent, 80 micron thick polyethylene tunnels. The average day (max) and night (minimum) temperatures during the cultivation season are shown in FIG. 6.

During winter in Israel, most days have full sunlight and only average of 82 rainy days from August to June, producing about 620 mm precipitation. The commercial fields are not treated with growth regulators and regularly are irrigated and fertilized according to need during fruit production.

Yield observations and fruit quality characteristics are averaged from 2 years of data collected from the 2006 to 2007 and 2007 to 2008 growing seasons. Flower measurements and characteristics are from secondary flowers unless otherwise noted. Fruit characteristics and measurements are from secondary fruit unless otherwise noted.

Color references are made to The Royal Horticultural Society Colour Chart (R.H.S.), except where general colors of ordinary significance are used. Color values were taken under daylight conditions between 10:00 a.m. to noon in Bet-Dagan, Israel. The approximate age of the observed plants is 4 to 6 months.

The following Tables 3 to 9 describe fruit, plant, stolon, foliage, fruiting truss, flower and pest/disease characteristics of the new strawberry ‘TAMIR’.

TABLE 3

FRUIT CHARACTERISTICS	
Characteristic	‘TAMIR’
Color of mature fruit	Orange-red, RHS 34A
Color of internal flesh	Orange-red, RHS 33B
Length (cm)	5 cm to 7 cm
Width (cm)	3.2 cm to 4 cm
Ratio length/width	Longer than broad
Calyx diameter (cm)	3.0 cm to 3.5 cm
Average weight (gm)	About 22 gram
Achene color	Yellow-green, RHS 154B
Number of achenes per berry	About 200
Achene weight (mg)	Not relevant
Marketable yield (gm/plt)	Monthly yield
	Nov. 53 gram/plant
	Dec. 143 gram/plant
	Jan. 155 gram/plant
	Feb. 184 gram/plant
	Mar. 232 gram/plant
	Apr. 165 gram/plant
	May 119 gram/plant
	Total: 1051 gram/plant
Size	Large
Predominant shape	Conical
Difference in shapes between primary and secondary fruit	Slight
Band without achenes	Narrow

TABLE 3-continued

FRUIT CHARACTERISTICS	
Characteristic	‘TAMIR’
Unevenness of surface	Weak
Evenness of color	Slightly uneven
Glossiness	Medium to strong
Insertion of achenes	Level with
Insertion of calyx	Level
Attitude of the calyx	Clasping or detached
Size of calyx in relation to fruit diameter	Same size
Adherence of calyx	Strong
Firmness of skin	Not relevant
Firmness of flesh	Firm
Distribution of red color of the flesh	When viewed in longitudinal section - marginal distribution of orange-red color of flesh of fruit (orange-red color (RHS 33B) towards outer rim of flesh of fruit, and white (RHS N155B) near inner part of flesh of fruit)
Hollow center expression	Weak
Flavor	Strong
Soluble solids (% brix)	8.5-10%
Time of first flowering	Very early, from the beginning of October
Time of first harvesting	Very early, from the beginning of November
Harvest period	November to June
Type of bearing	Partially remontant

TABLE 4

PLANT CHARACTERISTICS	
Characteristic	‘TAMIR’
Height (cm)	8 cm to 13 cm
Spread (cm)	28 cm to 33 cm
Size	Medium
Habit	Flattened - globular
Density	Sparse.
Vigor	Moderate

TABLE 5

STOLON CHARACTERISTICS	
Characteristic	‘TAMIR’
Average number per plant	Few
Fruiting field	Few, until the first harvest
Average Length:	About 2.5 m
Color:	Green, RHS 144B
Anthocyanin coloration	Greyed-red RHS 181C
Anthocyanin intensity	Very weak to weak
Diameter at bract (mm)	3 mm to 4 mm
Pubescence	Strong

TABLE 6

FOLIAGE CHARACTERISTICS	
Characteristic	‘TAMIR’
Foliage:	
Color of upper surface	Medium to dark green, RHS 137A
Color of under side	Light green, RHS 138B
Shape in cross section	Slightly concave
Interveinal blistering	Weak

TABLE 6-continued

FOLIAGE CHARACTERISTICS	
Characteristic	‘TAMIR’
Glossiness	Weak to medium
Number of leaflets	Only three
Terminal Leaflet:	
Length (cm)	6 cm to 8 cm
Width (cm)	About 6 cm
Length/width ratio	Slightly longer than broad
Incision of margin	Crenate
Shape of base	Rounded
Shape of teeth	Rounded
Petiole:	
Length (cm)	8 cm to 9 cm
Diameter (mm)	About 3 mm
Petiolule length (mm)	4 mm to 7 mm
Pubescence	Dense
Attitude of hairs	Outwards
Stipule:	
Length (mm)	About 30 mm
Width (mm)	About 10 mm
Anthocyanin coloration	None
Color	Yellow-green, RHS 145B

TABLE 7

FRUITING TRUSS CHARACTERISTICS	
Characteristic	‘TAMIR’
Length (cm)	About 6 cm
Position relative to foliage	Level with
Pubescence	Strong
Anthocyanin intensity	None
Attitude at first pick	Prostrate
Difference between primary and secondary flower	Very slight

TABLE 8

FLOWER CHARACTERISTICS	
Characteristic	‘TAMIR’
Petal color	
Mature (upper)	White, RHS 155B
Mature (lower)	White, RHS 155B
Immature (upper)	Not relevant
Immature (lower)	Not relevant
Petal shape	
Overall	Rounded
Apex	Rounded
Base	Rounded
Petal length (mm)	10 mm to 12 mm
Petal width (mm)	11 mm to 13 mm
Petal length/width ratio	Slightly broader than long
Number of petals/flower	Typically 5, sometimes 6
Sepals color	
Mature (upper)	Medium green, RHS 137C
Mature (lower)	Light green, RHS 138B
Immature (upper)	Not relevant
Immature (lower)	Not relevant
Sepal shape	
Overall	Elliptic
Apex	Typically acute

TABLE 8-continued

FLOWER CHARACTERISTICS	
Characteristic	‘TAMIR’
Base	Truncate
Sepal length (mm)	10 mm to 12 mm
Sepal width (mm)	4 mm to 6 mm
Sepal length/width ratio	Longer than broad
Number of sepals/flower	Typically 10, sometimes 20
Corolla diameter (mm)	30 mm to 34 mm
Calyx diameter (mm)	35 mm to 40 mm
Size of calyx relative to corolla	Slightly longer
Size of inner calyx relative to outer calyx	Slightly smaller
Relative position of petals	Overlapping

TABLE 9

REPRODUCTIVE ORGANS	
Characteristic	‘TAMIR’
Androecium:	
Stamen:	
Number per flower:	About 20 to 30.
Filament:	
Length:	About 3 mm to 4 mm.
Color:	Light yellow green, closest to RHS 145D.
Anther:	
Length:	About 1 mm.
Color:	Yellow-orange, RHS 17A.
Gynoecium:	
Receptacles:	
Shape:	Spherical.
Length:	About 5 mm to 6 mm.
Diameter:	About 4 mm to 6 mm.
Color:	Green, RHS 143C.
Style:	
Length:	About 3 mm to 4 mm.
Color:	Dark yellow, closest to RHS 13A.

TABLE 10

PEST AND DISEASE REACTIONS	
Characteristic	‘TAMIR’
Two spotted spider mite	No testing/observations
Lygus bug	No testing/observations
Flower thrips	No testing/observations
Powdery mildew	Tolerant
Botrytis fruit rot	Moderately tolerant
Angular leaf spot	No testing/observations

We claim:

1. A new and distinct strawberry plant named ‘TAMIR’, as herein described and illustrated by the characteristics set forth above, having a low number of daughter plants in the propagation nursery, not requiring chilling for flower induction, and having very early fruit production from the first week of November.

FIG. 1



FIG. 2

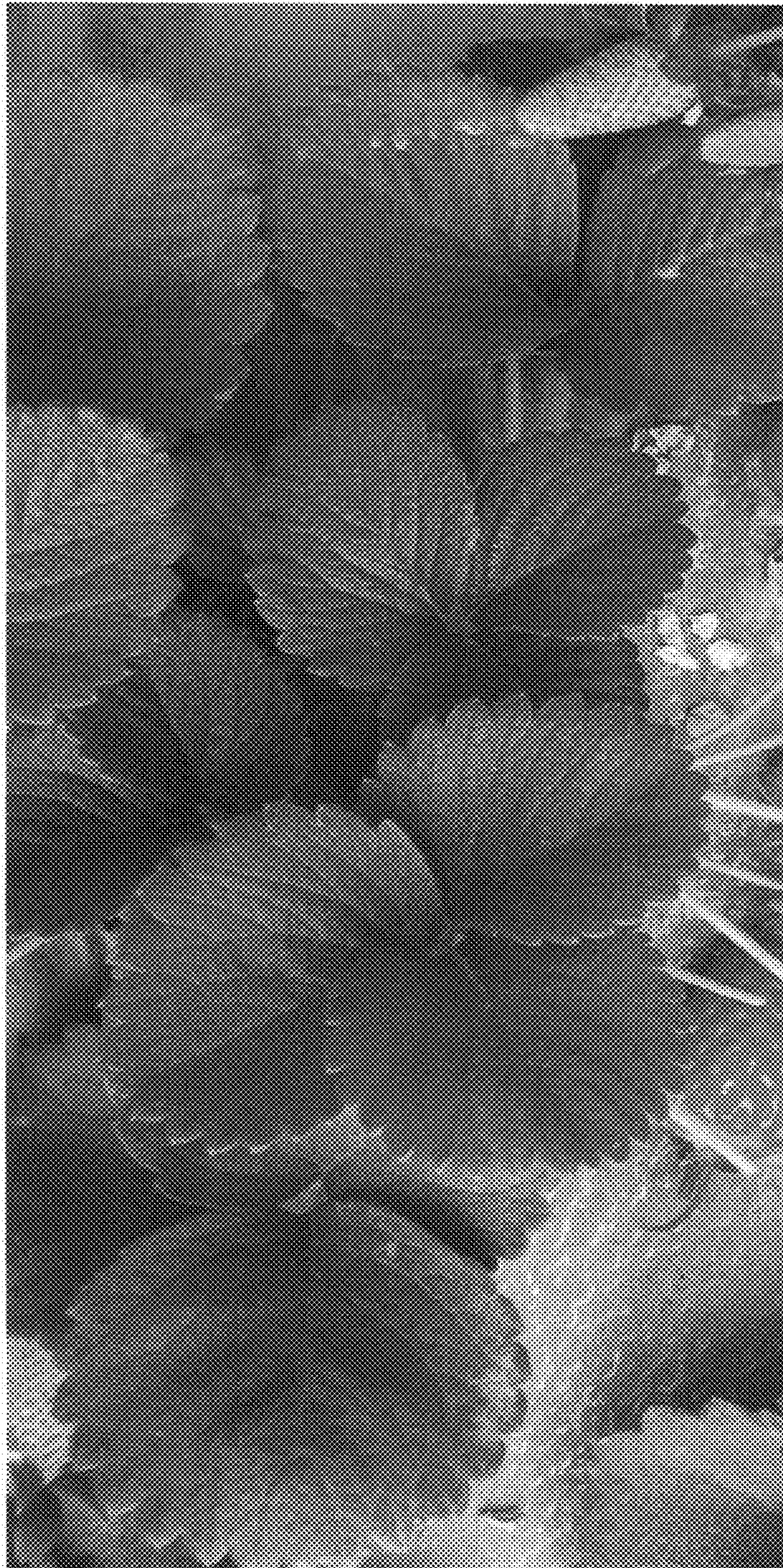


FIG. 3A

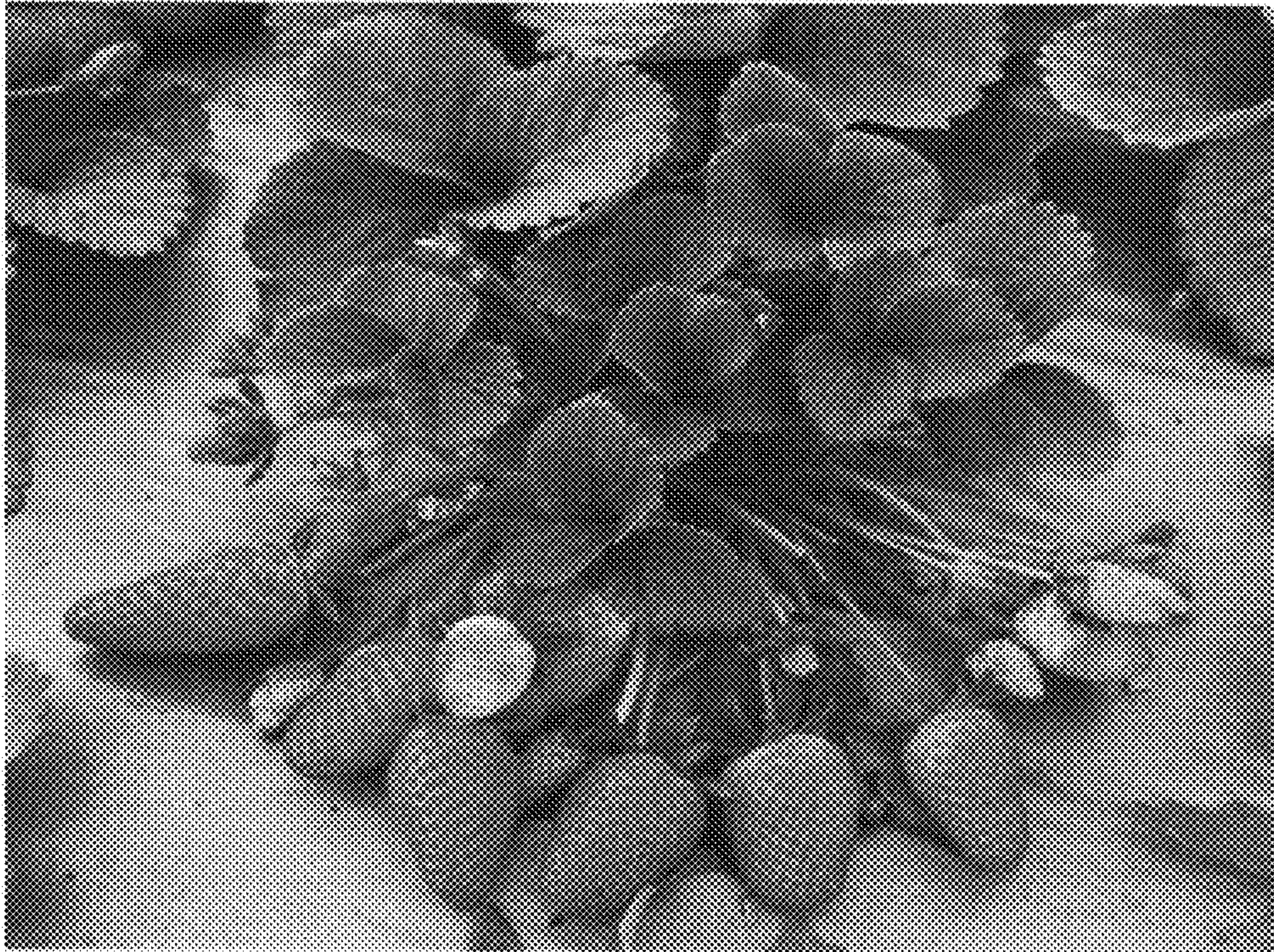


FIG. 3B



FIG. 4

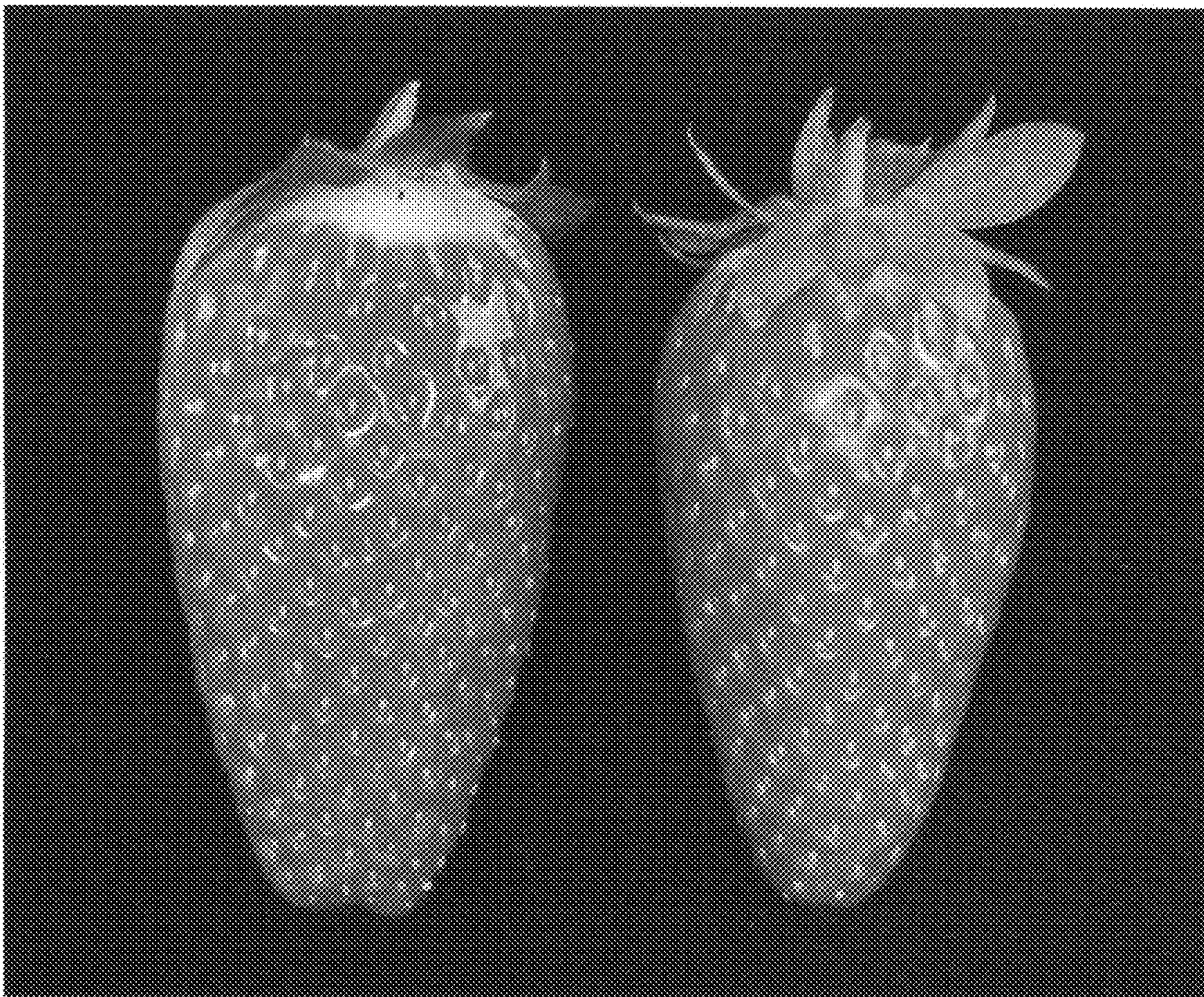


FIG. 5

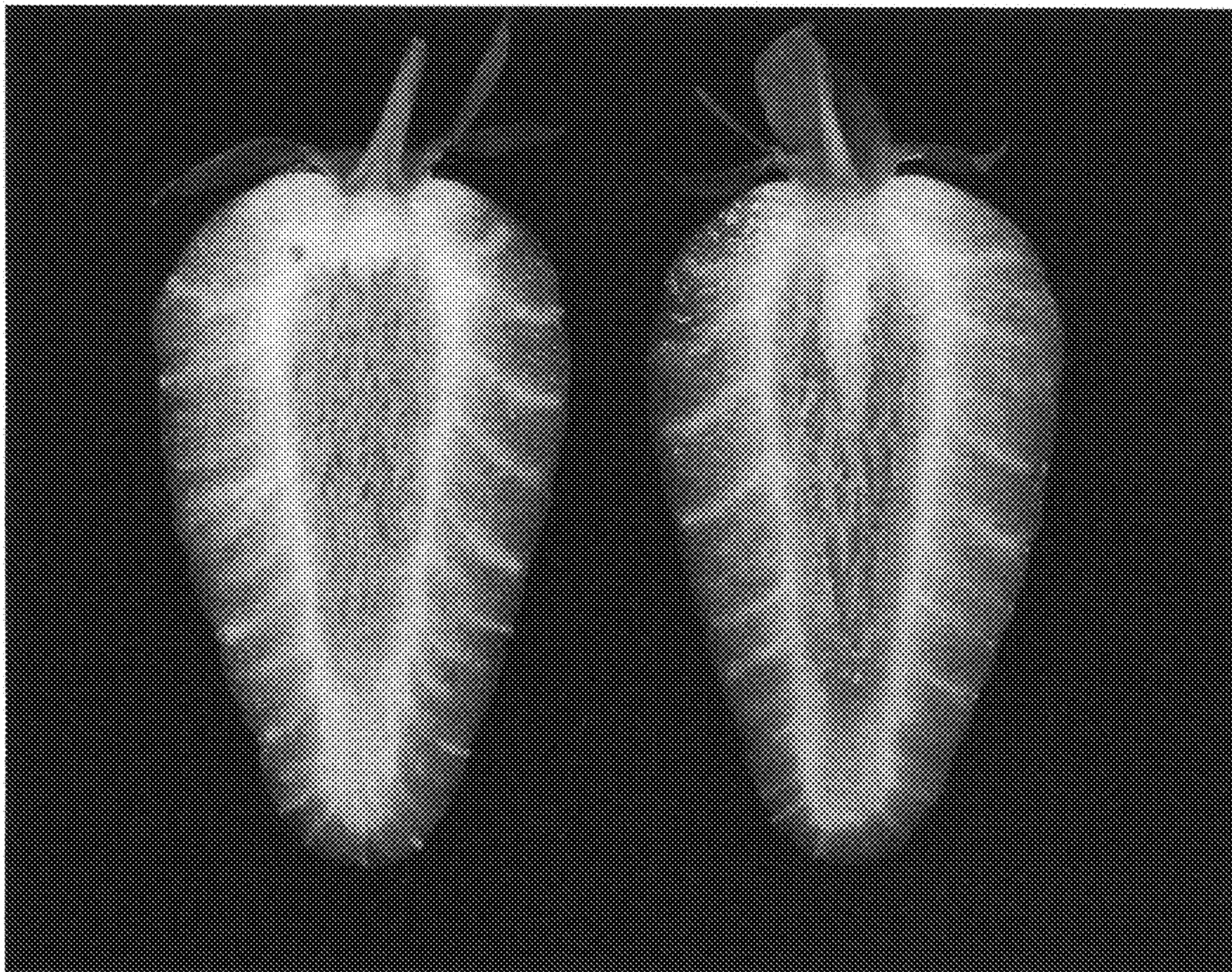


FIG. 6

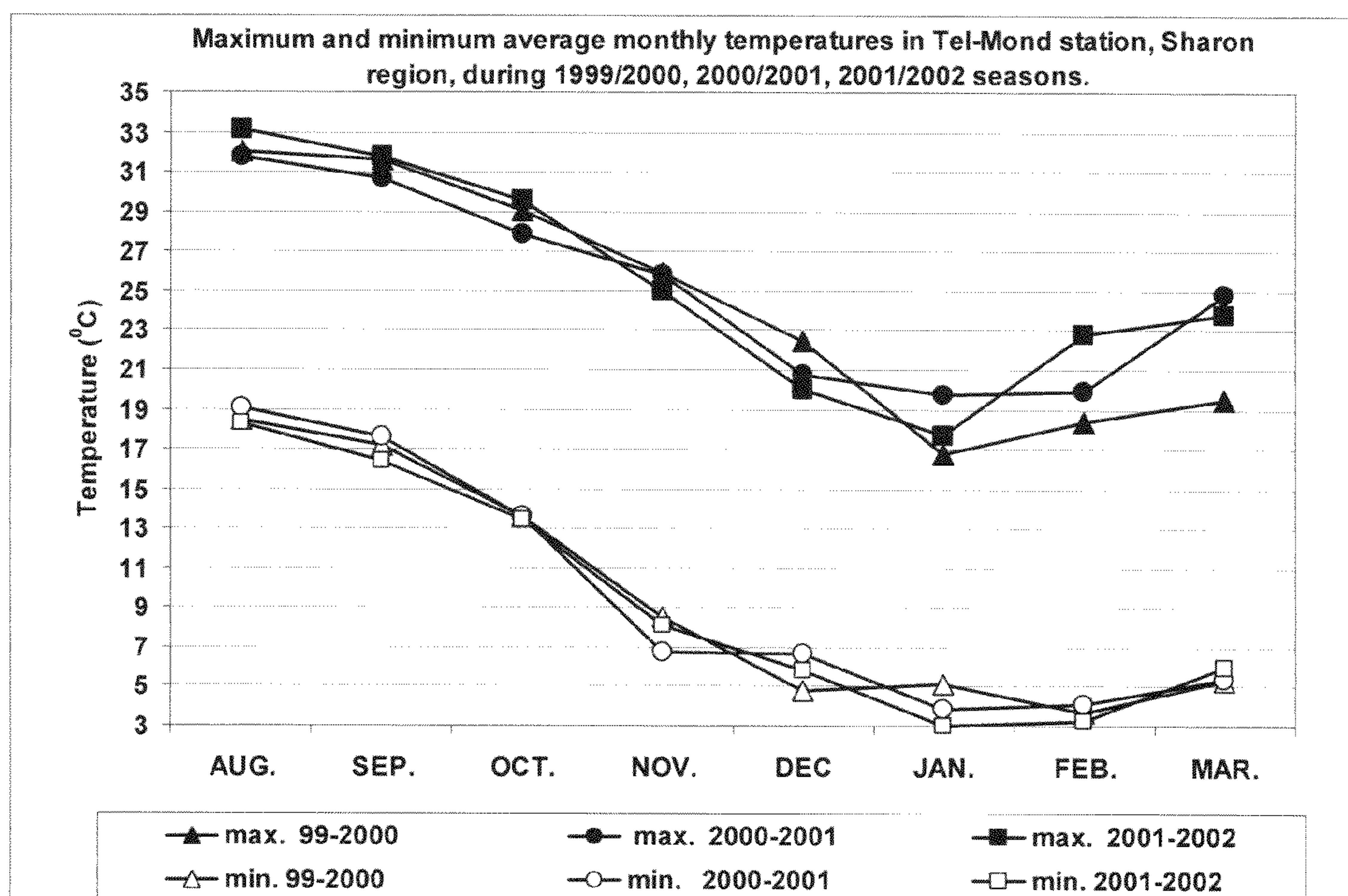


FIG. 7

